

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Home Healthcare & Diagnostic Center Management System

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, interface, and operational requirements for the Home Healthcare & Diagnostic Center Management System (HHDMS). It serves as the primary reference for all stakeholders including business owners, software architects, developers, QA engineers, and project managers.

This document governs the design, development, testing, and delivery of the HHDMS platform and represents the agreed-upon scope of the system.

1.2 Scope

The HHDMS is a centralized, multi-role digital platform that manages the complete lifecycle of home healthcare service delivery, including:

- Multi-specialty doctor consultations (MBBS first-call + 11 specialist categories)
- Adult and Pediatric Nursing services
- Portable Diagnostic services (USG/Sonography and X-Ray)
- Nutritionist consultations (home visit and teleconsultation)
- Caregiver assignments (Male, Female, Child; Day/Night/24-Hour/Respite)
- Call Center operations with live queue management
- Real-time GPS tracking of all field personnel
- Patient registration, medical records, and prescription management
- Scheduling, dispatch, billing, payments, and financial reporting
- SMS, WhatsApp, and push notification communications

1.3 System Users & Roles

Role	Type	Key Responsibilities	Platform Access
Call Center Agent	Internal Staff	Incoming call handling, patient registration, appointment booking, ticket management	Web (Desktop)
MBBS Doctor (First Call)	Medical Professional	First assessment, vital recording, test ordering, specialist referral	Web + Mobile App
Specialist Doctor (11 types)	Medical Professional	Detailed diagnosis, treatment plan, specialist report, follow-up	Web + Mobile App
Adult Nurse	Medical Professional	Vital monitoring, medication, wound care, shift reports	Mobile App

Role	Type	Key Responsibilities	Platform Access
Pediatric Nurse	Medical Professional	Pediatric care, growth monitoring, vaccination records	Mobile App
Nutritionist	Medical Professional	Diet assessments, customized diet charts, follow-ups	Web + Mobile App
Caregiver (Male/Female/Child)	Support Staff	Daily patient activities, shift check-in/out, condition reporting	Mobile App
Sonologist	Diagnostic Staff	USG image capture, structured report generation	Mobile App + Device
X-Ray Technician	Diagnostic Staff	X-Ray capture, technical report, radiation compliance	Mobile App + Device
System Administrator	Internal Staff	Full system access, user management, configuration, reports	Web (Admin Panel)
Patient / Guardian	External User	View records, track provider, receive notifications	Patient Portal (Web/App)

1.4 System Users & Roles

Term / Acronym	Definition
HHOMS	Home Healthcare & Diagnostic Center Management System
MRN	Medical Record Number — unique identifier assigned to each patient
MBBS	Bachelor of Medicine and Bachelor of Surgery — general practitioner degree
USG	Ultrasonography — portable ultrasound scanning service
DICOM	Digital Imaging and Communications in Medicine — medical imaging standard
GPS	Global Positioning System — used for real-time field staff tracking
RBAC	Role-Based Access Control — security model for access management
ETA	Estimated Time of Arrival — calculated for provider dispatched to patient
SRS	Software Requirements Specification
AHT	Average Handle Time — call center performance metric
RPO	Recovery Point Objective — maximum acceptable data loss window
RTO	Recovery Time Objective — maximum acceptable downtime
SMS	Short Message Service — text notification channel
API	Application Programming Interface — system integration layer
BMI	Body Mass Index — nutritional assessment metric
MFA	Multi-Factor Authentication — security login mechanism
SLA	Service Level Agreement — performance commitments

Term / Acronym	Definition
SpO2	Peripheral Oxygen Saturation — vital sign measurement
BP	Blood Pressure — vital sign measurement
QR	Quick Response code — used on lab requisition slips

2. SYSTEM OVERVIEW & ARCHITECTURE

2.1 High Level System Architecture

The HDDMS follows a three-tier architecture with a clear separation between presentation, business logic, and data layers. The system is cloud-hosted, mobile-first for field staff, and supports both online and offline operation.

Layer	Components	Technology Recommendations
Presentation Layer	Admin Web Dashboard, Patient Portal, Field Staff Mobile App, Call Center Interface	React.js / Next.js (Web), React Native (Mobile)
API / Business Logic Layer	REST API services, Authentication, Notification Engine, Scheduling Engine, GPS Engine	Python (Django/FastAPI), GraphQL optional
Data Layer	Relational DB (patient/operational data), Object Storage (DICOM images), Cache, Message Queue	PostgreSQL, AWS S3 / Cloudflare R2, Redis, RabbitMQ
Integration Layer	SMS Gateway, WhatsApp Business API, Payment Gateway, Google Maps API, DICOM Viewer	Third-party API integrations via secure webhooks / REST
Security Layer	API Gateway, WAF, MFA, Encryption at rest & in transit, RBAC engine	AWS API Gateway / Nginx, SSL/TLS, AES-256, JWT/OAuth2

2.2 Core Workflow Overview

All services within the HDDMS follow a unified workflow pattern:

Step	Action	System Actor	Key Data Generated
1	Patient/Guardian calls the service center	Call Center Agent	Caller ID, Patient Profile, Service Ticket
2	Agent registers patient and books service	Call Center Module	MRN, Appointment ID, SMS Confirmation
3	System notifies assigned provider	Notification Engine	Push/SMS Alert to Provider
4	Provider departs; live GPS tracking activated	GPS/Tracking Module	Real-time Location, ETA

Step	Action	System Actor	Key Data Generated
5	Patient receives live tracking link	Notification Engine	Web Tracking URL (no app needed)
6	Provider arrives; geofence auto check-in	GPS Module	Arrival Timestamp, GPS Coordinates
7	Service delivered; records entered	Service-specific Module	Consultation Notes, Vitals, Reports
8	Provider marks completion; GPS stops	GPS Module	Completion Timestamp, Visit Duration
9	Report/prescription sent to patient and referring doctor	Notification Engine	PDF Report, SMS/WhatsApp
10	Invoice generated; payment processed	Billing Module	Invoice, Payment Receipt, Commission Record

3. FUNCTIONAL REQUIREMENTS

The Call Center Module is the primary patient intake point of the system. It handles all inbound communications, patient registration, service booking, and urgent escalations.

3.1 Call Center Module

3.1.1 Caller Identification & Patient Lookup

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CC-001	Incoming Call Popup	Auto-display caller ID and retrieve matching patient records from the database the moment a call is received. If a patient is not found, prompt for new registration.	Must Have	Call Center Agent, Telecom API	Caller phone number, Telecom webhook event	Patient record popup / New registration prompt
CC-002	New Patient Registration	Register new patients with full demographic information. Mandatory fields: Full Name (Bengali/English), Phone Number (primary + emergency), Address (Division, District, Thana, Road), Date of Birth, Gender. Optional: Email, NID/Passport, Photo.	Must Have	Call Center Agent	Demographic inputs	MRN created, record saved, confirmation SMS

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CC-006	Unique Service Ticket	Auto-generate a unique alphanumeric Ticket ID (e.g. TKT-2026-00421) for every service request. Ticket tracks the complete patient journey from intake to billing.	Must Have	System (Auto)	Service booking event	Ticket ID, QR code, status trail

3.1.2 Appointment Booking

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CC-003	Multi-Service Booking	Book any of the 10+ service types from a single interface: MBBS Doctor, Specialist (choose from 11), Adult Nurse, Pediatric Nurse, Nutritionist, Caregiver (M/F/Child), USG, X-Ray.	Must Have	Call Center Agent	Service type, preferred date/time, address, special notes	Appointment record, provider assigned, SMS confirmation
CC-008	Real-Time Provider Availability	Display live availability of all providers filtered by service type, date, time slot, and geographic area. Show provider name, rating, and ETA to patient location.	Must Have	System, GPS Module	Service type, patient location, requested date	Available provider list with slots
CC-007	SMS Confirmation	Auto-send SMS to patient upon booking confirmation containing: Ticket ID, provider name and photo link, appointment date/time, and live tracking URL.	Must Have	SMS Gateway	Confirmed appointment record	SMS sent to patient and guardian (if registered)

3.1.3 Call Management

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CC-004	Call Log Management	Record every call with: Timestamp, Duration, Agent ID,	Must Have	Call Center	Telecom CDR data, agent input	Call log entry, daily call

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		Caller Number, Outcome (Booked / Enquiry / Escalated / Abandoned / Callback), linked Ticket ID.		Agent, System		summary report
CC-009	Call Queue Management	Display live queue with caller position, wait time estimate, and auto-callback option if agent unavailable. Queue dashboard visible to supervisor.	Must Have	System, Telecom API	Incoming call events	Queue display, wait time estimate, callback scheduled
CC-012	Call Recording & Playback	Record all inbound/outbound calls. Recordings stored for 90 days, accessible by supervisor only. Linked to patient Ticket ID for audit purposes.	Must Have	Telecom API, Cloud Storage	Audio stream from call	Stored recording with metadata, playback URL
CC-011	WhatsApp Integration	Accept service requests, send confirmations, share tracking links, and receive patient messages via WhatsApp Business API. Maintain conversation history per patient.	Should Have	WhatsApp Business API	Patient WhatsApp message	Reply, booking, or escalation action

3.1.4 Emergency Escalation

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CC-005	Urgent Call Forward	One-click escalation of active call to on-duty MBBS doctor with real-time transfer of patient records and call details.	Must Have	Call Center Agent, MBBS Doctor	Agent escalation trigger	Call transferred, doctor notified, ticket updated
CC-010	Emergency Alert Protocol	If a patient describes emergency symptoms (chest pain, difficulty breathing, loss of consciousness, etc.), the system flags the	Must Have	System, Call Center Agent	Emergency flag trigger	EMERGENCY ticket, MBBS alert, rapid dispatch

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		ticket as EMERGENCY, alerts on-duty MBBS, and initiates the fastest dispatch protocol.				

3.2 MBBS Doctor Module - First call (MB)

The MBBS Doctor (First Call) performs the initial patient assessment, records clinical data, orders diagnostics, and coordinates specialist referrals. This module is the clinical gateway of the system.

3.2.1 Assessment & Clinical Documentation

ID	Feature	Description	Priority	Actors	Inputs	Outputs
MB-001	Assignment Notification	Notify assigned MBBS doctor of new appointment via push notification, SMS, and in-app alert. Include patient name, address, appointment time, and call center notes.	Must Have	System, Notification Engine	Confirmed appointment event	Push notification + SMS to doctor
MB-002	Patient Information View	Display full patient profile: demographics, complete medical history, current medications, known allergies, past appointments, and linked call center notes.	Must Have	MBBS Doctor	Doctor opens patient record	Comprehensive patient summary view
MB-003	Vital Signs Recording	Structured vital signs form: Blood Pressure (Systolic/Diastolic mmHg), Pulse (bpm), Temperature (°C/°F), SpO2 (%), Respiratory Rate (breaths/min), Weight (kg), Height (cm). Auto-flag out-of-range values.	Must Have	MBBS Doctor	Vital signs measured at patient location	Stored vitals record with timestamp and alert if abnormal
MB-004	Preliminary Diagnosis	Enter chief complaint, history of present illness, review of systems, examination findings, and	Must Have	MBBS Doctor	Clinical findings	Diagnosis record linked to patient visit

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		preliminary diagnosis. Supports ICD-10 code lookup.				
MB-013	Differential Diagnosis Tool	AI-assisted differential diagnosis suggestions based on entered symptoms and vital signs. Doctor reviews and selects appropriate diagnosis. Tool is advisory only.	Should Have	System (AI Engine), MBBS Doctor	Symptom inputs, vitals	Ranked differential diagnosis list

3.2.2 Test Ordering & Lab Integration

ID	Feature	Description	Priority	Actors	Inputs	Outputs
MB-005	Diagnostic Test Ordering	Order blood tests, urine tests, ECG, and other diagnostics from a pre-configured test catalog. Supports test bundling (e.g. CBC + CMP). Tests routed to appropriate lab teams.	Must Have	MBBS Doctor	Selected tests, patient MRN	Test order record, lab team notification
MB-006	Lab Requisition Slip	Auto-generate a standardized, printable lab requisition slip containing: Patient name + MRN, Test names, Ordering doctor, Date/time, unique barcode/QR code for sample tracking.	Must Have	System	Confirmed test order	PDF requisition slip, barcode generated
MB-007	Lab Notification	Auto-notify patient (SMS/WhatsApp) and lab team (push/SMS) with test order details, collection instructions, and expected turnaround time.	Must Have	Notification Engine	Test order confirmed	Patient SMS, Lab team alert
MB-008	Test Results View	Display completed lab results with reference ranges, trend graphs for repeat tests, and critical value	Must Have	System, Lab Integration	Lab results uploaded	Results view with reference ranges and critical alerts

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		highlights. Notify the doctor immediately when critical values are received.				

3.2.3 Referral & Prescription

ID	Feature	Description	Priority	Actors	Inputs	Outputs
MB-009	Specialist Referral	Refer patients to one or more of 11 specialists. The system checks specialist availability by area/date and sends a referral package: MBBS notes + vitals + test results.	Must Have	MBBS Doctor	Selected specialist, referral notes	Referral record, specialist notified, appointment auto-offered
MB-010	Referral Chain Tracking	Maintain and display the complete care chain: Call Center → MBBS → Tests → Specialist → Follow-up. All actors can view the chain relevant to their role.	Must Have	System	Each workflow step completion	Timeline view of patient care journey
MB-011	Prescription Generation	Generate digital prescriptions with: Medication name (generic + brand), Dosage, Frequency, Duration, Route, Special instructions (with/without food, etc.).	Must Have	MBBS Doctor	Prescription inputs	Digital prescription, printable PDF
MB-012	Digital Signature	Apply doctor's registered digital signature (uploaded image + credential verification) on prescriptions and referral letters. Compliant with Bangladesh Medical Practice guidelines.	Must Have	MBBS Doctor, System	Doctor digital signature on file	Signed prescription/referral PDF
MB-014	Emergency Case Flag	One-click emergency flag on any patient record. Triggers priority specialist	Must Have	MBBS Doctor	Emergency flag trigger	Priority ticket, admin notified, specialist dispatched

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		dispatch, admin alert, and emergency notation on all linked records.				

3.3 Specialist Doctor Module (SP)

The HHDMs supports 11 specialist categories: Pulmonology (Lung), Cardiology (Heart), Neurology (Brain), Nephrology (Kidney), Dermatology (Skin), ENT, General Surgery, Gynecology & Obstetrics, Internal Medicine, Pain Management, and Oncology.

ID	Feature	Description	Priority	Actors	Inputs	Outputs
SP-001	Referral Notification	Notify specialists when patient referral reports are available. Include: MBBS referral notes, test results, imaging links, and urgency level.	Must Have	System, Notification Engine	Completed test results + MBBS referral confirmed	Push/SMS alert to specialist
SP-002	Complete Referral View	Display full referral chain: Call Center intake → MBBS assessment + vitals → Lab results with ranges → USG/X-Ray imaging (DICOM viewer). All in a chronological timeline.	Must Have	Specialist Doctor	Opens referral record	Integrated patient care timeline
SP-003	Visit or Teleconsultation	Book home visit (with GPS dispatch) or initiate video/audio teleconsultation directly from the specialist's availability calendar.	Must Have	Specialist Doctor, System	Appointment type selection	Scheduled appointment, patient notified
SP-004	Medical History View	Access complete patient history: chronic conditions, past hospitalizations, current medications, known allergies, family history, and previous specialist consultations.	Must Have	Specialist Doctor	Opens patient profile	Comprehensive medical history display
SP-005	Imaging Annotation	View USG and X-Ray images in embedded DICOM-compatible viewer. Annotate	Must Have	Specialist Doctor	Opens imaging study	Annotated DICOM image set saved to patient record

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		images with arrows, measurements, and text labels. Annotations saved with the report.				
SP-006	Specialist Diagnosis & Plan	Document detailed diagnosis (ICD-10), treatment plan, procedures recommended, prognosis, and patient instructions using specialty-specific templates.	Must Have	Specialist Doctor	Clinical assessment inputs	Structured specialist consultation report
SP-007	Specialist Prescription	Generate specialist prescriptions with drug interactions to check against the current medication list. Support for complex regimens, tapers, and conditional instructions.	Must Have	Specialist Doctor	Prescription inputs + current medications	Digital prescription with interaction warnings
SP-008	Consultation Report	Generate a structured specialist consultation report: Summary, Findings, Diagnosis, Treatment Plan, Prescription, Follow-up plan. Shareable as PDF via SMS/WhatsApp.	Must Have	System, Specialist Doctor	Completed consultation	PDF report delivered to patient and MBBS doctor
SP-009	Follow-up Scheduling	Schedule follow-up consultations directly from the visit completion screen. Auto-reminders sent 24 hrs and 1 hr before the follow-up.	Must Have	Specialist Doctor, System	Follow-up date/type selected	Appointment record, reminder schedule set
SP-010	Additional Test Request	Request additional diagnostics from within the specialist module. Follows the same lab ordering workflow as MBBS with specialist identity on the requisition.	Must Have	Specialist Doctor	Additional test selected	Test order, lab notified, requisition generated

ID	Feature	Description	Priority	Actors	Inputs	Outputs
SP-011	Specialty Templates	Pre-configured consultation templates for common conditions per specialty (e.g. Cardiac Stress Protocol, Neurological Examination, Renal Function Assessment).	Must Have	Specialist Doctor	Template selected	Pre-filled consultation form
SP-012	Availability Calendar	Visual calendar showing available home visit and teleconsultation slots. Specialists set their own availability; admin can override for emergency cases.	Must Have	Specialist Doctor, Admin	Calendar view requested	Available/block ed slot display
SP-013	Geographic Service Area	Specialist defines serviceable areas (by Division/District/Thana or custom polygon on map). The system only shows specialists as available for patients within their defined area.	Must Have	Specialist Doctor	Service area configuration	Filtered availability in booking system
SP-014	Consultation Fee Display	Auto-calculate and display consultation fee based on: service type (home visit/tele), specialist category, time of day (business hours / emergency surcharge), distance band.	Must Have	System	Appointment type and specialist profile	Fee displayed before booking confirmation

3.4 Nurse Module (NS)

Adult Nurses provide clinical care for patients 18 years and above. Pediatric Nurses specialize in care for patients under 18 years, including neonates and infants.

ID	Feature	Description	Priority	Actors	Inputs	Outputs
NS-001	Nurse Booking & Assignment	Book Adult or Pediatric nurse. System auto-selects by patient age, nurse availability, skill match, and proximity.	Must Have	Call Center Agent, System	Patient age, service requirements, date/time	Nurse assigned, patient confirmed, nurse notified

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		Admin can manually override.				
NS-003	Assigned Patient Schedule	Nurse's daily/weekly view of assigned patients with: Patient name, address, time slot, service requirements, and navigation link.	Must Have	Nurse	Opens schedule	Daily task list with map integration
NS-004	Vital Signs Recording	Record vitals at each visit: BP, Pulse, Temp, SpO2, RR, Blood Glucose (if required). Auto-compare with my previous visit. Alert if deterioration detected.	Must Have	Nurse	Measured values	Vital signs record with trend comparison
NS-005	Medication Administration	Log each medication administered: drug name, dose, route, time. Generates medication administration record (MAR) per patient per visit.	Must Have	Nurse	Medication details	MAR entry, timestamp, dose confirmation
NS-006	IV Fluid Monitoring	Track IV fluid administration: fluid type, rate (mL/hr), volume given, site condition, hourly fluid balance. Alert if rate deviates from order.	Must Have	Nurse	IV fluid order, monitoring inputs	Fluid balance record, deviation alert
NS-007	Wound Care Documentation	Upload geo-tagged, timestamped wound photos. Record wound measurements, condition, dressing applied, and healing progress across visits.	Must Have	Nurse, Mobile Camera	Wound photos, assessment inputs	Wound care record with photos, progress chart
NS-008	Nursing Care Report	Auto-generate structured nursing visit report on completion: vitals, medications, procedures, patient response, handover notes.	Must Have	System, Nurse	Completed visit data	PDF nursing report shared with referring doctor and patient

ID	Feature	Description	Priority	Actors	Inputs	Outputs
NS-010	Shift Handover Notes	Structured handover form for shift change: Current status, active concerns, medications due, physician orders, and patient/family instructions. Accepts nurse-to-nurse digital sign-off.	Must Have	Nurse	Shift end trigger	Handover document, receiving nurse notified
NS-011	Live GPS Tracking	Real-time GPS tracking active from shift start to shift end (see GPS Module 3.8).	Must Have	System, GPS	Shift start event	Live location visible to admin and patient
NS-012	Doctor Consultation Request	Nurses can initiate urgent consultation requests to MBBS or specialists if patient condition changes. Includes current vitals, concern summary, and patient record access.	Must Have	Nurse, System	Consultation request triggered	Doctor alert, response time tracked
NS-013	Nursing Supply Tracking	Track consumables used per patient per visit: gloves, dressings, syringes, IV lines, etc. Feeds into inventory module for replenishment planning.	Should Have	Nurse	Supply usage inputs	Inventory usage record, replenishment alert
NS-014	Pediatric Specific Care	Pediatric module additions: feeding logs (type, volume, frequency), growth monitoring (weight, height, head circumference vs WHO growth charts), vaccination tracking.	Must Have	Pediatric Nurse	Feeding and growth inputs	Growth chart, vaccination schedule, feeding log

3.5 Nutritionist Module (NU)

ID	Feature	Description	Priority	Actors	Inputs	Outputs
NU-001	Consultation Booking	Book nutritionist for home visit or	Must Have	Call Center Agent	Patient details,	Appointment booked,

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		teleconsultation. The system checks availability by area and displays the next available slot.			consultation type, date/time	nutritionist notified
NU-002	Medical Context Review	A nutritionist views a patient's medical conditions, active medications, allergies, lab results (kidney function, glucose, lipids), and specialist notes before consultation.	Must Have	Nutritionist	Opens patient record	Integrated medical summary relevant to nutrition
NU-003	Anthropometric Recording	Record height (cm), weight (kg), waist circumference (cm), hip circumference (cm). Auto-calculate BMI, BMI category, ideal body weight, and caloric needs.	Must Have	Nutritionist	Measured values	Anthropometric record, BMI category, caloric target
NU-004	Customized Diet Chart	Create personalized diet charts by meal (Breakfast, Mid-Morning, Lunch, Afternoon, Dinner, Bedtime) with specific foods, quantities, and preparation guidance.	Must Have	Nutritionist	Patient profile, food preferences, clinical needs	Structured diet chart document
NU-005	Condition-Specific Templates	Pre-built templates for: Diabetes (Type 1 & 2), CKD (Stage 3–5), Heart Disease, Hypertension, Obesity, Malnutrition, Post-surgical, Cancer, Pediatric malnutrition.	Must Have	Nutritionist	Condition selected	Pre-populated diet framework for customization
NU-006	Bilingual Diet Plan PDF	Generate print-ready PDF diet plan in both Bengali and English. Patient's name, date, caloric target, and nutritionist's signature included.	Must Have	System	Finalized diet chart	Bengali/English PDF, delivered via SMS/WhatsApp
NU-007	Follow-up Scheduling	Book follow-up at fixed intervals (2 weeks, 1 month, 3 months).	Must Have	Nutritionist, System	Follow-up date selected	Appointment created, reminder sent

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		Auto-reminders to patient.				
NU-008	Dietary Adherence Tracking	At each follow-up, record patient adherence score, challenges, and plan modifications. Track weight trend over time.	Should Have	Nutritionist	Follow-up assessment inputs	Adherence chart, weight trend, plan revision record
NU-009	Nutrient Calculation Tools	Built-in calorie and macronutrient calculator. Database of common Bangladeshi foods with calories, protein, carbs, fat, fiber per 100g. Exchange list for substitutions.	Must Have	Nutritionist, System	Food selections and quantities	Nutritional breakdown of meal plan
NU-010	Patient Education Materials	Upload and share PDF/image educational materials: portion guides, label reading, food groups, condition-specific handouts (Bengali and English).	Should Have	Nutritionist	Material uploaded	Shared with patient via portal and WhatsApp

3.6 Caregiver Module (CG)

Caregivers provide non-clinical daily living support. The module supports Male, Female, and Child patient caregivers, with four service duration types: Day Care (8 hrs), Night Care (8 hrs overnight), 24-Hour Care, and Respite Care.

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CG-001	Caregiver Booking	Book caregiver by: gender preference (Male/Female), patient type (Adult/Child/Elderly), care duration type, special skills (dementia care, post-surgical, pediatric), and availability.	Must Have	Call Center Agent	Patient needs, preferences, date/time	Caregiver assigned, patient and caregiver confirmed
CG-003	Service Type Management	Support four service durations: Day Care (7am–3pm), Night Care (10pm–6am),	Must Have	System, Admin	Service type selection	Shift schedule, caregiver assignment, billing type

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		24-Hour Care (rotational), Respite Care (temporary relief for family caregivers, minimum 4 hrs).				
CG-005	Daily Activity Logging	Log daily care activities: personal hygiene (bathing, grooming), mobility assistance, feeding assistance, oral medication administration, companionship, exercise.	Must Have	Caregiver	Activity completion inputs	Daily activity log per patient
CG-006	GPS Shift Check-In/Out	Caregiver checks in when within 100m of patient address (geofence verified). Check-out verified the same way. Timestamps used for attendance and billing.	Must Have	GPS Module, Caregiver	Arrival at patient address	Verified check-in/out, attendance record
CG-007	Condition Change Reporting	Caregiver reports any observed changes: patient fall, medication refusal, behavioural changes, physical symptoms. Instantly alerts the assigned nurse and MBBS doctor.	Must Have	Caregiver, Notification Engine	Caregiver submits report	Alert to nurse/doctor, incident record created
CG-008	Attendance & Timesheets	Auto-generate daily and monthly timesheets from GPS-verified check-in/out records. Exported to payroll system.	Must Have	System	Check-in/out records	Timesheet, payroll export
CG-009	Caregiver Rating & Feedback	After each shift, the patient or guardian rates the caregiver (1–5 stars) and submits written feedback. Average ratings displayed on caregiver profile.	Must Have	Patient/Guardian	Post-shift feedback submission	Rating updated on profile, trends tracked

ID	Feature	Description	Priority	Actors	Inputs	Outputs
CG-010	Training Records	Maintain caregiver training log: training date, subject, certificate expiry. Alert admin 30 days before certificate expiry.	Should Have	Admin, System	Training record inputs	Training history, expiry alert
CG-011	Specialized Caregiver Booking	Book caregivers with specific certifications: Dementia Care, Post-Surgical Recovery, Pediatric Care, Bedridden Patient Care, Terminal/Palliative Care.	Must Have	Call Center Agent	Specialty requirement selected	Matched certified caregiver assigned

3.7 Diagnostic Services Module

3.7.1 USG / Sonography Service (US) — Sonologist

Portable USG service dispatched to the patient's home with a qualified Sonologist and DICOM-capable portable ultrasound machine.

ID	Feature	Description	Priority	Actors	Inputs	Outputs
US-001	USG Booking	Book USG specifying body part and scan type: Abdomen, Pelvis, Obstetric (with GA), Thyroid, Breast, Scrotum, MSK, Vascular Doppler, Echo (bedside). Include referral doctor details.	Must Have	Call Center Agent	Body part, scan type, referral doctor, patient details	USG appointment, requisition generated
US-002	Sonologist Assignment	Auto-assign sonologist based on: availability, geographic proximity, scan specialty, and equipment readiness status.	Must Have	System, Scheduling Module	Appointment details, sonologist availability	Assignment confirmed, sonologist notified
US-003	Referral Details View	Sonologist views: referring doctor's clinical notes, patient history relevant to scan, any prior USG studies for comparison.	Must Have	Sonologist	Opens scan request	Clinical context for scan

ID	Feature	Description	Priority	Actors	Inputs	Outputs
US-005	Scan Findings & Measurements	Structured report entry: organ dimensions (in cm), echogenicity, vascularity, lesion characterization (size, margins, composition), fetal biometry (if OB). Standard measurement tools built in.	Must Have	Sonologist	Scan observations	Structured measurement record
US-006	USG Report Generation	Auto-generate structured USG report: Patient info, scan type, clinical indication, technique, findings (organ by organ), impression, and recommendations. Includes representative images.	Must Have	System, Sonologist	Completed findings and captured images	PDF USG report
US-007	DICOM Image Storage	All captured images stored in DICOM-compliant format with metadata. Linked to patient MRN and report. Accessible from referring doctor and specialist module.	Must Have	DICOM Storage, System	Image capture from portable machine	DICOM study stored, accessible via viewer
US-008	Specialist Image Annotation	Referring specialist views and annotates USG images using embedded DICOM viewer: distance/area measurements, text labels, comparison side-by-side.	Must Have	Specialist Doctor	Opens imaging study	Annotated images saved to report
US-009	Report Delivery Notification	Auto-notify referring doctor (push + email) and patient (SMS + WhatsApp) when USG report is finalized. PDF downloadable from patient portal.	Must Have	Notification Engine	Report finalized event	Notification sent, report on portal

ID	Feature	Description	Priority	Actors	Inputs	Outputs
US-010	Equipment Maintenance Log	Track calibration dates, last service, fault reports, and next scheduled maintenance for each portable USG machine. Alert admin 7 days before calibration due.	Should Have	Admin, Sonologist	Maintenance record inputs	Equipment health log, maintenance alert

3.7.2 Portable X-Ray Service (XR) — X-Ray Technician

ID	Feature	Description	Priority	Actors	Inputs	Outputs
XR-001	X-Ray Booking	Book portable X-Ray specifying anatomical region and projection views: AP, PA, Lateral, Oblique, Decubitus. Include referring doctor details and clinical indication.	Must Have	Call Center Agent	Body part, views required, clinical indication	X-Ray appointment, radiation consent form
XR-002	Technician Assignment	Auto-assign X-Ray technician based on availability, proximity, and equipment readiness. Manual override by admin.	Must Have	System, Admin	Appointment details, technician availability	Assignment confirmed, technician notified
XR-003	Patient Precaution Check	System prompts and records safety check before dispatch: Pregnancy status (female patients), metal implants, pacemaker, allergy to contrast. Adds precaution notes to technician's assignment.	Must Have	Call Center Agent, System	Patient demographic and medical history	Safety check recorded, precaution notes dispatched
XR-005	Exposure Parameters Record	Record technical parameters for each projection: kVp, mAs, SID, grid use, film/detector size. Stored for radiation audit.	Must Have	X-Ray Technician	Technical parameters	Exposure record stored
XR-006	Technical Report	Technician enters brief technical assessment: image	Must Have	X-Ray Technician	Technical assessment inputs	Technical notes attached to study

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		quality (diagnostic/non-diagnostic), positioning adequacy, any artifacts, repeat exposures.				
XR-007	DICOM Image Storage	All X-Ray images stored in DICOM format with EXIF-equivalent metadata (patient MRN, date, view, kVp/mAs). Linked to patient record.	Must Have	DICOM Storage	Image digitization from DR/CR plate	DICOM study stored
XR-008	Radiologist Interpretation	Consulting radiologist/specialist accesses images via DICOM viewer, enters formal radiological interpretation report, and digitally signs it.	Must Have	Radiologist/Specialist	Opens X-Ray study	Signed radiological report
XR-009	Report Delivery	Auto-notify referring doctor and patient when X-Ray interpretation is complete. PDF report delivered via SMS/WhatsApp and available on patient portal.	Must Have	Notification Engine	Interpretation completed event	Report delivery notification
XR-010	Radiation Safety Compliance	Track cumulative radiation exposure per patient (for repeat X-Rays). Flag if dose limits approached. Log technician radiation badge readings monthly.	Must Have	System, Admin	Exposure records, badge readings	Dose log, safety flag if threshold approached
XR-011	Machine Maintenance Log	Track calibration dates, service history, and compliance certificates for each portable X-Ray unit. Alert admin 14 days before calibration due.	Should Have	Admin	Maintenance record inputs	Equipment maintenance log, alert

3.8 Live GPS Tracking & Dispatch Module(LT)

The GPS module provides real-time location visibility of all field staff (Doctors, Nurses, Caregivers, Sonologists, X-Ray Technicians) to administrators, call center agents, and patients. Privacy is enforced: tracking is active only during assigned duty hours.

3.8.1 Real-Time Tracking

ID	Feature	Description	Priority	Actors	Inputs	Outputs
LT-001	GPS Location Capture	Capture device GPS coordinates every 30 seconds when field staff is on active duty. Accuracy target: ≤ 10 meter radius. Uses device GPS + network triangulation fallback.	Must Have	Field Staff Mobile App, GPS Hardware	Active duty status, device location	GPS coordinates + timestamp stored every 30 seconds
LT-002	Admin Live Map Dashboard	Single-view map showing all active field staff in real-time. Color-coded by role (doctor/nurse/caregiver/diagnostic). Clickable pin shows: name, role, current patient, ETA, status.	Must Have	Admin, Call Center Supervisor	Opens live dashboard	Real-time map with all active staff pins
LT-004	ETA Calculation	Calculate ETA from field staff current location to patient address using Google Maps / OpenStreetMap real-time traffic data. Update dynamically as traffic changes.	Must Have	System, Google Maps API	Staff location, patient address	Calculated ETA, updated in real time
LT-005	Patient Tracking Link	Generate a unique, time-limited web URL sent to the patient by SMS/WhatsApp when the provider is dispatched. The patient opens a link in any browser — no app required.	Must Have	System	Dispatch event triggered	Unique tracking URL, live map for patient
LT-016	Offline GPS Caching	Mobile app caches GPS readings when internet is unavailable. Auto-syncs cached data when connection is restored. Ensures no location data gap	Must Have	Field Staff Mobile App	No internet connection	Cached GPS data, synced on reconnect

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		in areas with poor connectivity.				
LT-022	Map Integration	Integrate Google Maps API for routing, ETA, and live tracking display. OpenStreetMap as fallback. Auto-route optimization for multi-stop routes.	Must Have	Google Maps API, OpenStreet Map	Route request	Optimized route displayed with turn-by-turn

3.8.2 Automated Visit Lifecycle Events

ID	Feature	Description	Priority	Actors	Inputs	Outputs
LT-008	Geofence Auto Check-In	Auto-trigger check-in event when the field staff device enters 100-meter radius of the patient's registered address. No manual action required.	Must Have	GPS Module	Device enters 100m geofence	Check-in event recorded, patient notified of arrival
LT-009	Near-Arrival Alert	Push notification + SMS to patient when provider is within 5 minutes of their location.	Must Have	System, Notification Engine	ETA ≤ 5 minutes	Patient notification: 'Your provider is 5 minutes away'
LT-012	Visit Status Timeline	Track and display complete visit lifecycle with timestamps: Dispatched → En Route → Near (5 min) → Arrived → Service Started → Service Completed.	Must Have	System, GPS Module	Each lifecycle event trigger	Timeline displayed on admin dashboard and patient portal
LT-010	Manual Location Confirmation	Field staff can tap the 'Reached Location' button for GPS-verified confirmation. System validates coordinates match patient address within tolerance.	Must Have	Field Staff Mobile App, GPS	Manual tap by field staff	GPS-verified arrival confirmation
LT-011	Service Start/Complete Timestamps	Field staff marks 'Service Started' and 'Service Completed'.	Must Have	Field Staff Mobile App, GPS	Manual trigger by field staff	Verified service

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		Both require GPS verification. Timestamps used for duration calculation and billing.				duration record
LT-020	Auto-Stop Tracking on Logout	GPS tracking automatically stops when field staff logs out or ends duty. Tracking link expires. Location data no longer captured.	Must Have	System	Logout / duty-end event	Tracking stopped, link expired
LT-023	Privacy Control	Live location shared with patients only for the duration of their specific visit — from dispatch to service completion. Location not shared at any other time.	Must Have	System	Visit start/end events	Time-limited location access enforced

3.8.3 Safety & Alert Features

ID	Feature	Description	Priority	Actors	Inputs	Outputs
LT-007	Route Deviation Alert	Alert admin and call center supervisor if field staff deviates more than 500 meters from the planned route for more than 5 minutes.	Must Have	GPS Module, Notification Engine	GPS coordinates vs planned route	Deviation alert to admin, logged for review
LT-013	Stationary Alert	Alert admin if field staff GPS shows stationary position for >30 minutes during an active visit (possible emergency or equipment issue).	Must Have	GPS Module, System	GPS shows no movement > 30 min	Admin alert, staff called to check status
LT-014	SOS Emergency Button	One-tap SOS button on mobile app. Instantly sends GPS coordinates, field staff identity, and SOS alert to admin dashboard and emergency contact. Triggers automated call-back.	Must Have	Field Staff Mobile App	SOS button pressed	Instant GPS alert to admin + emergency contact, call initiated

ID	Feature	Description	Priority	Actors	Inputs	Outputs
LT-021	ETA Delay Notification	Auto-notify patients if ETA changes by more than 15 minutes due to traffic, route change, or prior appointment overrun.	Must Have	System	ETA change > 15 min detected	Patient SMS/WhatsApp: updated ETA + apology message

3.8.4 Operational & Audit Features

ID	Feature	Description	Priority	Actors	Inputs	Outputs
LT-006	Route History Log	Store complete GPS route taken for each home visit. Accessible for: quality review, dispute resolution, and payroll verification.	Must Have	System, Admin	Visit completion	Route log stored, viewable as map replay
LT-015	Location History for Payroll	Generate monthly location history per field staff for payroll validation: visits completed, hours logged, GPS-verified attendance.	Must Have	System, Admin	Payroll period end	Location-verified attendance report
LT-018	Distance-Based Reimbursement	Calculate total distance traveled per staff member per day using GPS route logs. Export to transport reimbursement calculation.	Must Have	System	End of duty day	Distance report per staff per day
LT-024	Daily Movement Report	Automated daily report per field staff: Locations visited, routes taken, idle times, total distance, duty hours. Available to admin.	Should Have	System	End of day trigger	Daily movement report PDF/CSV
LT-017	All-Staff Live Map (Admin)	Admin dashboard view of all active staff simultaneously on a single map. Filter by role, district, availability status.	Must Have	Admin	Opens live staff map	Full fleet visibility on map

3.9 Patient Management(PM)

ID	Feature	Description	Priority	Actors	Inputs	Outputs
PM-001	Centralized Patient Database	Single source of truth for all patient records across all service types. Accessible by all authorized roles with RBAC filtering.	Must Have	System, All Roles	Patient registration event	Unified patient profile
PM-002	Unique MRN	Assign unique Medical Record Number (format: MRN-YYYY-XXXXXX) at registration. Used as primary identifier across all modules, lab requisitions, invoices, and reports.	Must Have	System	Patient registration	MRN generated, printed on all documents
PM-003	Demographics & Contacts	Store: Full name (Bengali + English), DOB, Age, Gender, NID, Blood Group, Address (full GPS coordinates), Primary phone, Alternative phone, Emergency contact (name + relationship + phone).	Must Have	Call Center Agent	Registration form	Patient record with all demographics
PM-004	Medical History	Store and update: Chronic conditions, Past surgical history, Known allergies (drug/food/environmental — severity noted), Current medications (name + dose + frequency), Family history.	Must Have	MBBS Doctor, Specialist	Medical history inputs	Complete, searchable medical history
PM-005	Service History	Complete chronological log of all services availed: date, service type, provider, diagnosis, prescription, and invoice. Filterable by service type or date range.	Must Have	System	Each completed service	Service history timeline
PM-006	Patient Portal	Patient/Guardian accesses: personal records, appointment history, lab results,	Must Have	Patient/Guardian	Portal login	Personalized health dashboard

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		imaging reports, prescriptions, invoices, live tracking link, and nutritionist diet plans.				
PM-007	Family Account	One guardian account linked to multiple family members (e.g. parent managing elderly parent + child records). Each member has a separate MRN; guardians can switch between them.	Must Have	System	Family account setup	Guardian dashboard with family member switcher
PM-009	Digital Consent Forms	Obtain and store digital consent before invasive services (IV insertion, X-Ray, USG with probe). Consent recorded with patient/guardian signature and timestamp.	Must Have	System, Call Center	Pre-appointment workflow	Signed digital consent stored, linked to appointment
PM-010	Feedback & Complaints	The patient submits service feedback (1–5 stars + comments) and formal complaints via portal or SMS link. Complaints routed to admin with 24-hour response SLA.	Must Have	Patient/ Guardian, Admin	Feedback submission	Feedback record, complaint ticket if rating ≤ 2 stars

3.10 Scheduling & Dispatch(SD)

ID	Feature	Description	Priority	Actors	Inputs	Outputs
SD-001	Centralized Scheduling Dashboard	Master scheduling view for all service types. Filter by date, provider type, district, and status. Drag-and-drop rescheduling on the calendar grid.	Must Have	Admin, Call Center	Opens scheduling dashboard	Full appointment calendar, all services
SD-003	Route Optimization	When multiple appointments are booked for the same provider on the same	Must Have	System, Google Maps API	Multiple appointments for one provider	Optimized visit sequence with total distance

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		day, the system auto-suggests optimal visit sequence based on geographic proximity.				
SD-004	Provider Reminders	Auto-SMS + push notification to provider 2 hours and 1 hour before scheduled appointment. Include patient name, address, contact number, and appointment details.	Must Have	Notification Engine	Time trigger: 2hr and 1hr before	Reminder notifications to provider
SD-005	Cancellation & Rescheduling	The patient or admin can cancel or reschedule. Cancellation policy enforced: free if >4 hrs before appointment; fee if <4 hrs. Reason mandatory. Provider notified immediately.	Must Have	System, Admin	Cancellation/re schedule request	Appointment status updated, provider and patient notified, fee applied if applicable
SD-006	Real-Time Status Tracking	All appointments display current status: Scheduled → Provider Notified → Dispatched → En Route → Arrived → In Progress → Completed → Cancelled / Rescheduled.	Must Have	System	Each status change event	Status visible on dashboard, patient portal, and provider app
SD-007	Recurring Appointments	Set up recurring schedules for long-term care: daily nursing visits, weekly specialist follow-ups, monthly nutritionist reviews. Auto-creates appointments for defined periods.	Must Have	Admin, Call Center	Recurrence configuration	Recurring appointment series, reminders automated
SD-008	Manual Admin Override	Admin can override any auto-scheduling decision: assign specific provider, change time slot, adjust geographic area constraints.	Must Have	Admin	Override trigger	Override recorded with reason, new assignment applied

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		Override reason logged.				

3.11 Billing & Payment(BP)

ID	Feature	Description	Priority	Actors	Inputs	Outputs
BP-001	Auto-Charge Calculation	Calculate charges per service based on: service type, provider level (MBBS / Specialist category), visit type (home/tele), time of day (business/emergency), distance band, visit duration.	Must Have	System	Service completion event	Itemized charge breakdown
BP-002	Itemized Invoice	Generate invoice showing: each service line with rate, add-ons (transport, after-hours surcharge), applicable taxes, discounts, total amount due. Invoice numbered (INV-YYYY-XXXXX).	Must Have	System	Service completion	PDF invoice, sent to patient via SMS/ WhatsApp
BP-003	Multiple Payment Methods	Accept: Cash (recorded manually), Debit/Credit Card (online gateway), Mobile Banking (bKash, Nagad, Rocket), Bank Transfer. Payment status updated in real time.	Must Have	System, Payment Gateway	Payment received	Payment record, transaction ID, receipt
BP-004	Payment Status Tracking	Track per invoice: Pending → Partially Paid → Paid → Refunded → Waived. Support partial payment with outstanding balance carried forward.	Must Have	System	Payment events	Invoice payment status, outstanding balance
BP-005	Digital Receipt	Auto-generate receipt (RCP-YYYY-XXXXX)	Must Have	System	Payment confirmed	Digital receipt PDF

ID	Feature	Description	Priority	Actors	Inputs	Outputs
		upon payment. PDF sent to patient via SMS/WhatsApp and stored on patient portal.				
BP-006	Insurance Documentation	Generate insurance claim documentation package: itemized invoice, consultation report, diagnosis (ICD-10), prescriptions, and lab results in a single PDF bundle.	Should Have	System, Admin	Insurance claim request	Claim documentation bundle PDF
BP-007	Provider Commission Tracking	Calculate and track commission per provider per service based on pre-configured commission rates. Monthly commission statement generated per provider.	Must Have	System	Invoice paid	Commission record, monthly statement per provider
BP-008	Financial Reports	Daily, weekly, monthly revenue reports: by service type, by provider, by district, by payment method. Exportable as Excel/PDF.	Must Have	System, Admin	Report period selected	Financial summary report
BP-009	Package Pricing	Pre-defined care packages (e.g. Post-Surgery 7-Day Care Package, Monthly Caregiver Package). Auto-apply package price when package booked. Track remaining sessions.	Should Have	Admin, System	Package booking event	Package record, session tracker, discounted invoice
BP-010	Discounts & Promotions	Apply percentage or fixed discounts. Promo codes with validity dates and usage limits. Discount reasons are mandatory. All discounts logged for financial audit.	Should Have	Admin	Discount/promo code applied	Discount recorded on invoice, promo usage tracked

3.12 Reporting & Analytics(RA)

ID	Feature	Description	Priority	Actors	Inputs	Outputs
RA-001	Daily Operational Summary	Auto-generated daily report: total appointments (by type), completed, cancelled, revenue collected, outstanding invoices, new patient registrations, active staff.	Must Have	System	End of day trigger	Daily operations dashboard + PDF report
RA-002	Doctor Performance Metrics	Per doctor: consultations completed, average rating, average response time from notification to departure, referrals made, prescriptions generated.	Must Have	System	Reporting period selected	Doctor performance scorecard
RA-003	Nurse/Caregiver Attendance	Attendance report using GPS-verified check-in/out: days worked, hours per shift, late check-ins, early check-outs, per staff member.	Must Have	System	Payroll period end	Attendance report with GPS verification flags
RA-004	Call Center KPIs	Calls answered, abandoned, average handle time (AHT), first-call resolution rate, bookings per agent, escalations initiated.	Must Have	System	Reporting period selected	Call center performance dashboard
RA-005	Patient Satisfaction	Aggregate patient feedback scores by service type, provider, and district. Trend over time. Flag providers with consistently low ratings.	Must Have	System	Reporting period selected	Satisfaction report with provider-level breakdowns
RA-006	Inventory Usage	Consumable usage by service type and provider. Stock level alerts when thresholds are reached.	Should Have	System, Admin	Ongoing	Inventory usage report, low-stock alerts

ID	Feature	Description	Priority	Actors	Inputs	Outputs
RA-007	Revenue Analytics	Revenue by service type, by district, by provider, by payment method, by package. Monthly/quarterly/annual trend charts.	Must Have	System, Admin	Reporting period selected	Revenue analytics dashboard with charts
RA-008	Custom Reports	User-configurable report builder: select dimensions (date range, service type, provider, district), metrics (revenue, volume, ratings), and output format (Excel, PDF).	Should Have	Admin	Custom report parameters	Generated report in selected format

3.13 Notification Engine(NF)

ID	Trigger Event	Recipient	Channel	Content
NF-001	Appointment confirmed	Patient/ Guardian	SMS	Ticket ID, provider name, date/time, tracking link
NF-002	24 hrs before appointment	Patient/ Guardian	SMS + WhatsApp	Appointment reminder with provider name and time
NF-002b	1 hr before appointment	Patient/ Guardian	SMS + WhatsApp	Appointment reminder with ETA
NF-003	New assignment created	Provider	Push + SMS	Patient name, address, service type, time
NF-004	Lab/imaging report finalized	Referring Doctor + Patient	Push + SMS + WhatsApp	Report available, download link
NF-005	Invoice generated	Patient/ Guardian	SMS + WhatsApp	Invoice total, payment link, due date
NF-005b	Invoice overdue 3 days	Patient/ Guardian	SMS	Payment reminder with outstanding balance
NF-006	WhatsApp message received	Call Center Agent	In-App Alert	Message preview, patient ID
NF-007	Emergency flag raised	Admin + On-Duty MBBS	Push + SMS + Email	EMERGENCY alert with patient details and location
NF-008	Provider dispatched	Patient/ Guardian	SMS + WhatsApp	Provider name, photo, ETA, live tracking link
NF-009	Provider arrives	Patient/ Guardian	Push + SMS	Provider has arrived notification

ID	Trigger Event	Recipient	Channel	Content
NF-010	Prescription generated	Patient/ Guardian	WhatsApp + Portal	Prescription PDF download link
NF-011	Diet plan generated	Patient/ Guardian	WhatsApp + Portal	Diet plan PDF (Bengali + English)
NF-012	Follow-up appointment due	Patient/ Guardian	SMS + WhatsApp	Reminder 48 hrs before follow-up

4. NON FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

Metric	Requirement	Measurement Method
Concurrent Users	Minimum 200 simultaneous users without performance degradation	Load testing (e.g. JMeter)
Page Load Time	< 3 seconds for 95th percentile of page loads under normal load	Real User Monitoring (RUM)
API Response Time	< 500ms for data queries; < 2 seconds for report generation	API monitoring tools
Caller ID Popup	< 2 seconds from call initiation to popup display	Automated test with simulated calls
GPS Update Latency	< 5 seconds from GPS capture to map display	GPS lag testing
Report Generation	< 10 seconds for standard reports; < 30 seconds for complex analytics	Automated report generation tests
Database Scalability	10,000 patients initial; upgradeable to 100,000+ without schema redesign	DB architecture review
File Storage	DICOM images ≤ 50MB per study; PDF reports ≤ 5MB	File size monitoring

4.2 Security Requirements

Requirement	Details
Encryption at Rest	AES-256 encryption for all PHI (Patient Health Information) stored in database and file storage
Encryption in Transit	TLS 1.3 for all API communications; HTTPS enforced on all web interfaces
Authentication	Username + password + OTP (SMS-based MFA) for admin and all medical professionals. Session token expiry: 15 minutes idle

Requirement	Details
Role-Based Access Control	Minimum 10 roles with granular permission sets. No cross-role data leakage. RBAC enforced at API layer.
Audit Logging	All PHI access, modifications, and deletions logged with: user ID, timestamp, IP address, action taken. Retained for 5 years.
Data Residency	All patient data stored on servers within Bangladesh or in a jurisdiction compliant with Bangladesh healthcare data laws
Session Management	Auto-logout after 15 minutes of inactivity. Single active session per user enforced.
API Security	API keys + JWT tokens. Rate limiting to prevent brute force. Input validation + parameterized queries to prevent SQL injection.
DICOM Security	Medical images accessible only through authenticated DICOM viewer with role-based access. Direct URL access blocked.
Mobile App Security	Certificate pinning, jailbreak/root detection, remote wipe capability for lost devices
Password Policy	Minimum 10 characters, 1 uppercase, 1 lowercase, 1 number, 1 special character. 90-day mandatory rotation for admin accounts.
Penetration Testing	Annual third-party penetration testing. Critical vulnerabilities remediated within 72 hours.

4.3 Availability & Reliability

Requirement	Target	Notes
System Uptime	99.5% per month	Excludes scheduled maintenance windows
Scheduled Maintenance	Maximum 4 hours per month	Between 2:00 AM – 5:00 AM only, advance notice to users
Recovery Time Objective (RTO)	< 4 hours for full system recovery after critical failure	Defined in Disaster Recovery Plan
Recovery Point Objective (RPO)	< 1 hour of data loss acceptable	Daily full backup + hourly incremental
Backup	Daily automated database backup to secondary location	Geo-redundant backup storage recommended
Failover	Auto-failover to hot standby within 5 minutes for database failure	Active-passive DB setup
GPS Offline Mode	Field staff app caches data offline; syncs when connection restored	Minimum 4 hours of cached data

Requirement	Target	Notes
Call Center Availability	Manual paper-based backup process defined for system outage	Business Continuity Plan (BCP) documented

4.4 Usability Requirements

- Bengali and English full support: all UI text, reports, diet plans, notifications, and error messages must be available in both languages
- Responsive design: fully functional on 5-inch mobile screens through 27-inch desktop monitors
- Call center interface: common tasks (incoming call popup → patient lookup → booking) completable in maximum 5 clicks / 2 minutes
- Field staff mobile app: optimized for one-handed use; large touch targets; offline-first design
- Patient portal: accessible to non-technical users; minimal medical jargon in patient-facing content
- Color contrast ratio: minimum 4.5:1 (WCAG 2.1 AA compliance for accessibility)
- Error messages: clear, actionable (not generic system errors); support both languages
- Training materials: in-app contextual help and video walkthrough for each role

4.4 Scalability Requirements

Dimension	Initial Capacity	Scale-Up Target
Registered Patients	10,000	100,000+ without performance impact
Active Service Providers	200	1,000+
Concurrent Users (Peak)	200	1,000+
Appointments per Day	500	5,000+
DICOM Storage	1 TB	50 TB+ (cloud-elastic storage)
SMS/WhatsApp Notifications	2,000/day	20,000+/day
Geographic Coverage	Single city (Dhaka)	All 64 districts of Bangladesh

5. DATA ENTITIES & DATA DICTIONARY

5.1 Core Data Entities

Entity	Key Attributes	Related Entities	Retention
Patient	MRN, Full Name (EN+BN), DOB, Gender, Blood Group, NID, Address (GPS), Primary Phone, Emergency Contact, Active Medications,	MedicalRecord, Appointment, Invoice, LabReport, ImagingReport, Prescription	Lifetime of patient + 10 years

Entity	Key Attributes	Related Entities	Retention
	Allergies, Chronic Conditions		
Doctor (MBBS)	Doctor ID, NID, BMDC Reg. No., Full Name, Phone, Email, Photo, Availability Schedule, Service Area (GPS polygon), Digital Signature	Appointment, MedicalRecord, Prescription, Referral, Commission	Duration of employment + 7 years
Specialist Doctor	Doctor ID, BMDC Reg. No., Specialty (enum: 11 types), Sub-specialties, Consultation Fee (home/tele), Service Area, Availability Calendar	Appointment, SpecialistReport, Prescription, Commission	Duration of employment + 7 years
Nurse	Nurse ID, Type (Adult/Pediatric), BNMC Reg. No., Qualifications, Skills, Shift Preference, GPS Device ID	Appointment, NursingReport, VitalSigns, InventoryLog	Duration of employment + 7 years
Caregiver	Caregiver ID, Gender, Experience (years), Specializations (enum), Verification Status, Training Certs, Rating	CaregiverShift, ActivityLog, Attendance, Commission	Duration of employment + 7 years
Nutritionist	Nutritionist ID, Qualifications, Specializations (enum: conditions), Availability, Service Area	Appointment, DietChart, FollowUpNote	Duration of employment + 7 years
Sonologist	Sonologist ID, Qualifications, Assigned Equipment ID(s), USG Specializations	Appointment, USGReport, ImagingStudy	Duration of employment + 7 years
X-Ray Technician	Tech ID, Qualifications, Radiation Safety Certificate (expiry), Assigned Machine ID(s)	Appointment, XRayReport, ImagingStudy, RadiationLog	Duration of employment + 7 years
Appointment	Appt ID, Ticket ID, Patient MRN, Service Type, Provider ID, Date/Time, Status (enum: 8 states), Address, Special Notes, Cancellation Reason	Patient, Provider, Invoice, ServiceTicket	5 years

Entity	Key Attributes	Related Entities	Retention
MedicalRecord	Record ID, Patient MRN, Visit Date, Provider ID, Chief Complaint, Vitals, Examination Findings, Diagnosis (ICD-10), Plan	Prescription, LabOrder, Referral, Patient	Lifetime + 10 years
LabReport	Report ID, Order ID, Patient MRN, Test Name, Result Value, Unit, Reference Range, Critical Flag, Lab Tech ID, Report Date	MedicalRecord, LabOrder, Patient	Lifetime + 10 years
ImagingStudy (DICOM)	Study ID, Patient MRN, Modality (USG/XR), Body Part, Study Date, Technician ID, DICOM Series UIDs, Storage URL	ImagingReport, MedicalRecord, Patient	Lifetime + 10 years
ImagingReport	Report ID, Study ID, Patient MRN, Findings, Impression, Annotated Images, Reporter ID, Digital Signature, Report Date	ImagingStudy, MedicalRecord, Patient	Lifetime + 10 years
Prescription	Rx ID, Patient MRN, Doctor ID, Visit Date, Medications (array: name, dose, frequency, duration, route, instructions), Digital Signature	MedicalRecord, Patient, Doctor	Lifetime + 10 years
Invoice	Invoice ID, Ticket ID, Patient MRN, Line Items (array), Subtotal, Discount, Tax, Total, Payment Status, Payment History	Appointment, Payment, Patient	10 years (financial records)
ServiceTicket	Ticket ID, Patient MRN, Opening Agent, Open DateTime, Service Types, Current Status, Timeline Log (array), Closing DateTime	Patient, Appointment, Invoice, Provider	5 years
GPSLog	Log ID, Provider ID, Appointment ID, Timestamp, Latitude,	Provider, Appointment	2 years (operational)

Entity	Key Attributes	Related Entities	Retention
	Longitude, Accuracy, Speed, Network Status		
CallLog	Call ID, Agent ID, Caller Number, Patient MRN (if matched), Start Time, Duration, Outcome (enum), Recording URL, Ticket ID (if created)	CallCenterAgent, Patient, ServiceTicket	90 days (recording); 2 years (metadata)

6. SYSTEM INTERFACES

6.1 User Interfaces

Interface	Platform	Primary Users	Key Design Requirements
Admin Web Dashboard	Web (Desktop primary)	System Admin, Supervisor	Full feature access, live maps, report dashboards, user management, override controls
Call Center Interface	Web (Desktop)	Call Center Agent	Popup-on-call, fast booking (<5 clicks), queue view, live staff availability, one-screen workflow
MBBS Doctor App	Android 8.0+ / iOS 13+ + Web	MBBS Doctor	Offline-capable, quick vitals entry, test ordering, referral workflow, digital signature
Specialist Doctor App	Android 8.0+ / iOS 13+ + Web	Specialist Doctor	DICOM viewer, annotation tools, referral chain view, availability calendar
Nurse/Caregiver App	Android 8.0+ / iOS 13+	Nurses, Caregivers	One-handed use, large buttons, offline vitals entry, photo upload, GPS tracking
Diagnostic Staff App	Android 8.0+ / iOS 13+	Sonologist, X-Ray Tech	DICOM integration, report templates, equipment logs
Patient Portal	Web + App (Android/iOS)	Patient, Guardian	Simple UI, record viewing, appointment history, live tracking, prescription download
Nutritionist Interface	Web + Android/iOS	Nutritionist	Diet chart builder, bilingual PDF export, food database lookup

6.2 Hardware Interfaces

Hardware	Integration Method	Data Format	Notes
Portable USG Machine	USB / WiFi / Bluetooth export	DICOM PS3	Technician reviews images on app before upload; DICOM metadata auto-populated

Hardware	Integration Method	Data Format	Notes
	from machine to app		
Portable X-Ray (DR/CR)	WiFi / USB transfer from detector/plate to app	DICOM PS3	Technical parameters (kVp/mAs) auto-read from machine where supported
Barcode / QR Scanner	Camera-based (mobile app) or USB scanner (lab/call center)	QR string → JSON	Lab requisition slip QR scanned for sample registration
GPS (Mobile Device)	Native device GPS API (Android/iOS) + Network triangulation fallback	Lat/Lon + accuracy (m)	Background GPS with battery optimization; offline caching enabled
Printer (Receipt/Reports)	AirPrint (iOS) / Google Cloud Print / PDF export to local printer	PDF	All printable documents generated as PDF; printing optional

6.3 Software/API Interfaces

Service	Provider (Recommended)	Integration Method	Data Exchanged
SMS Gateway	SSL Wireless / Twilio / BD local provider	REST API + Webhook	Outbound SMS: appointment alerts, OTPs, reminders. Inbound: keyword responses
WhatsApp Business API	Meta WhatsApp Business API (via BSP)	REST API + Webhook	Outbound: confirmations, reports, tracking links. Inbound: booking requests, queries
Payment Gateway	SSLCommerz / bKash for Business / Nagad for Business	REST API + IPN callback	Payment initiation, status callback, refund, receipt
Google Maps Platform	Google Maps API	REST API + JavaScript SDK	Geocoding, routing, ETA, live tracking display, geofencing
OpenStreetMap	OpenStreetMap + OSRM routing	REST API	Fallback mapping and routing if Google Maps unavailable
Cloud Storage (DICOM)	AWS S3 / Cloudflare R2	S3-compatible API	DICOM study upload/download, signed URL access for viewer
Email Service	SendGrid / AWS SES	REST API / SMTP	System alerts, report delivery, admin notifications
Push Notifications	Firebase Cloud Messaging (FCM)	REST API	Push alerts to Android/iOS mobile app users

Service	Provider (Recommended)	Integration Method	Data Exchanged
Video Consultation	Daily.co / Whereby / Jitsi (self-hosted)	JavaScript SDK + API	Video/audio stream for teleconsultation; recording optional

7. TECHNOLOGY STACK RECOMMENDATION

Layer	Recommended Technology	Justification
Frontend (Web)	React.js/Next.js	SEO-ready, SSR for fast initial load, large ecosystem, component reuse
Frontend (Mobile)	React Native	Cross-platform iOS + Android from single codebase; GPS, camera, offline support
Backend API	Python (FastAPI)	High concurrency for real-time GPS; async I/O; REST + WebSocket support
Database (Primary)	PostgreSQL 16+/ MongoDB(NO SQL)	ACID compliance for medical records; JSON support for flexible fields; geo-spatial extensions
Cache Layer	Redis	Session storage, real-time GPS feed caching, notification queue
Message Queue	RabbitMQ / Amazon SQS	Decoupled notification dispatch; SMS/WhatsApp queue management
File Storage	AWS S3 or Cloudflare R2	DICOM image storage; scalable, durable, geo-redundant
DICOM Server	Orthanc (open source)	DICOM PS3 compliance, REST API, viewer integration, self-hosted
Real-time Communication	Socket.IO / WebSocket	Live GPS map updates, call center live queue, real-time status changes
Hosting	AWS / Azure (BD region or Singapore)	Auto-scaling, managed services, compliance, low latency to BD
CI/CD	GitHub Actions + Docker + Kubernetes	Automated testing, containerized deployment, auto-scaling
Monitoring	Datadog / New Relic + Sentry	Performance monitoring, error tracking, uptime alerts

8. RISK REGISTER & MITIGATION

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-001	GPS accuracy insufficient in high-density urban areas or buildings	Medium	High	Network triangulation fallback; manual check-in option; geofence tolerance 150m in dense areas

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-002	DICOM machine not compatible with app integration	Medium	High	Pre-purchase compatibility validation; support universal DICOM export; image upload fallback
R-003	Bangladesh healthcare data privacy law changes	Low	High	Monitor DGDA and MoH guidelines; modular data storage design for easy compliance updates
R-004	Field staff resistance to GPS tracking (privacy concerns)	High	Medium	Clear communication: tracking only during duty; legal agreement; 'tracking off' auto on logout
R-005	SMS/WhatsApp delivery failures in remote areas	Medium	Medium	Multiple gateway redundancy; delivery status monitoring; in-app notification as fallback
R-006	Payment gateway downtime (bKash/Nagad outages)	Low	Medium	Multiple gateway integration; manual cash recording as immediate fallback
R-007	Medical data breach / unauthorized access	Low	Critical	AES-256 encryption, MFA, penetration testing, audit logs, access monitoring, incident response plan
R-008	Scope creep leading to delivery delays	High	High	MoSCoW prioritization enforced; change request process with impact assessment; phased delivery
R-009	Call center staff adoption challenges	Medium	High	UX-focused design (<5 clicks); phased training; dedicated super-user per team
R-010	Internet connectivity for field staff in poor-coverage areas	High	Medium	Offline mode with local data caching; auto-sync on reconnection; WiFi hotspot provision for critical staff

9. GLOSSARY & APPENDICES

9.1 Specialist Categories - Reference Table

Code	Specialty	Medical Field	Common Conditions Managed
SP-LUNG	Pulmonology	Respiratory Medicine	COPD, Asthma, Pneumonia, Pleural Effusion, Lung Cancer screening
SP-HEART	Cardiology	Cardiovascular Medicine	Hypertension, Heart Failure, Arrhythmia, Post-MI care, Angina
SP-BRAIN	Neurology	Nervous System	Stroke, Epilepsy, Parkinson's, Dementia, Peripheral Neuropathy
SP-KIDNEY	Nephrology	Renal Medicine	CKD, Dialysis management, Glomerulonephritis, Hypertensive Nephropathy

Code	Specialty	Medical Field	Common Conditions Managed
SP-SKIN	Dermatology	Skin & Hair	Eczema, Psoriasis, Wound care, Fungal infections, Skin cancer screening
SP-ENT	ENT	Ear, Nose & Throat	Sinusitis, Hearing loss, Vertigo, Voice disorders, Nasal polyps
SP-SURG	General Surgery	Surgical Care	Post-surgical wound care, Colostomy management, Drain management
SP-GYNAE	Gynecology & Obstetrics	Women's Health	Antenatal care, Postpartum care, Menstrual disorders, PCOS
SP-MED	Internal Medicine	General Adult Medicine	Diabetes, Complex multi-system conditions, Elderly care
SP-PAIN	Pain Management	Pain Medicine	Chronic pain, Cancer pain, Neuropathic pain, Palliative care
SP-ONCO	Oncology	Cancer Care	Chemotherapy support, Palliative care, Post-treatment monitoring

9.2 Requirements Priority Definitions(MoSCoW)

Must Have	Non-negotiable requirement. The system cannot launch without it.	System fails to meet core business objectives or legal requirements
Should Have	Important requirement. High value but the system can launch without it.	Significant inconvenience or workaround required; planned for near-term release
Could Have	Nice to have. Low impact if excluded from current scope.	Minor inconvenience; planned for a future release
Won't Have (this time)	Explicitly excluded from current scope. May be revisited.	No impact on current release; documented for future consideration